The Narrative Paradox
(is Still Alive)

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Uruguay



América Invertida
by Joaquín Torres García

2020–2023: My first years in research



Guarani

- ★ one of the few indigenous languages (worldwide) that continues to be spoken on a massive scale every day
- ★ 10 million speakers
- ★ an official language of MERCOSUR
- ★ …still considered a *low-resource language* in NLP



Low-resource vicious circle

2020-2023: My first years in research

We collected Guarani text from the internet

We trained word embeddings and used them to train Guarani-Spanish machine translation models

We met other researchers along the way and joined forces to create an even **bigger** dataset (still tiny in relative terms)

We had a meeting with one of the team members behind Google Translate

(In the days before the conference, Guarani-Spanish was included in Google Translate👀)



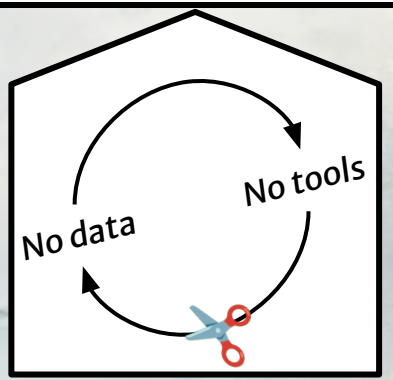
2020-2023: My first years in research

LONG STORY SHORT

- Program grammars
- Generate synthetic text
- Pre-train machine translation models on synthetic text
- Keep training them with high-quality data



In this corpus, every sentence is long... how can we get shorter, more basic Guarani sentences?



Theme track award

NAACL 2024

Grammar-based Data Augmentation for Low-Resource Languages: The Case of Guarani-Spanish Neural Machine Translation

Agustín Lucas, Alexis Baladón Ferreira de Araujo, Victoria Pardiñas, Marvin M. Agüero-Torales, Santiago Góngora, Luis Chiruzzo

This paper proposes a non-extractive method for augmenting a dataset on the native Latinamerican language: Guaraní. The method is automatic augmentation based on gramatical knowledge of guaraní. This offers a promising alternative to traditional methods like back-translation, especially in low-resource settings. By demonstrating the relevance of grammar-focused machine translation research, the paper provides valuable inspiration for researchers showing that grammars are still relevant in this setting.





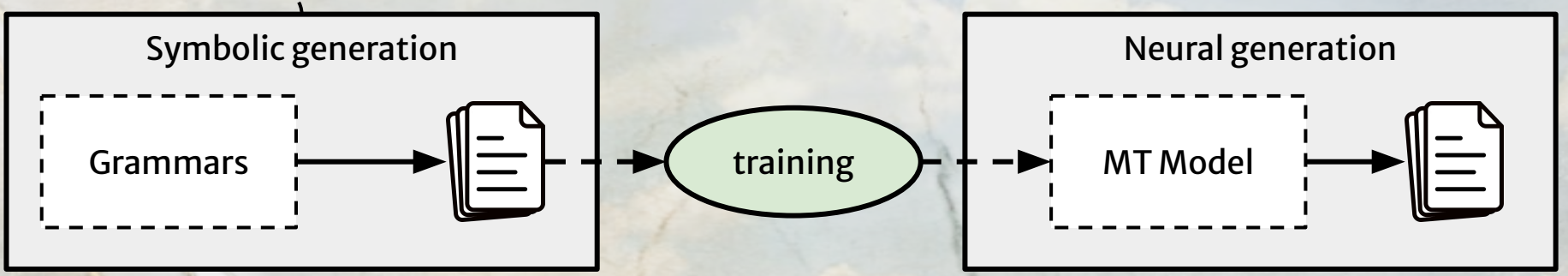
In this corpus, every sentence is long... how can we get shorter, more basic Guarani sentences?



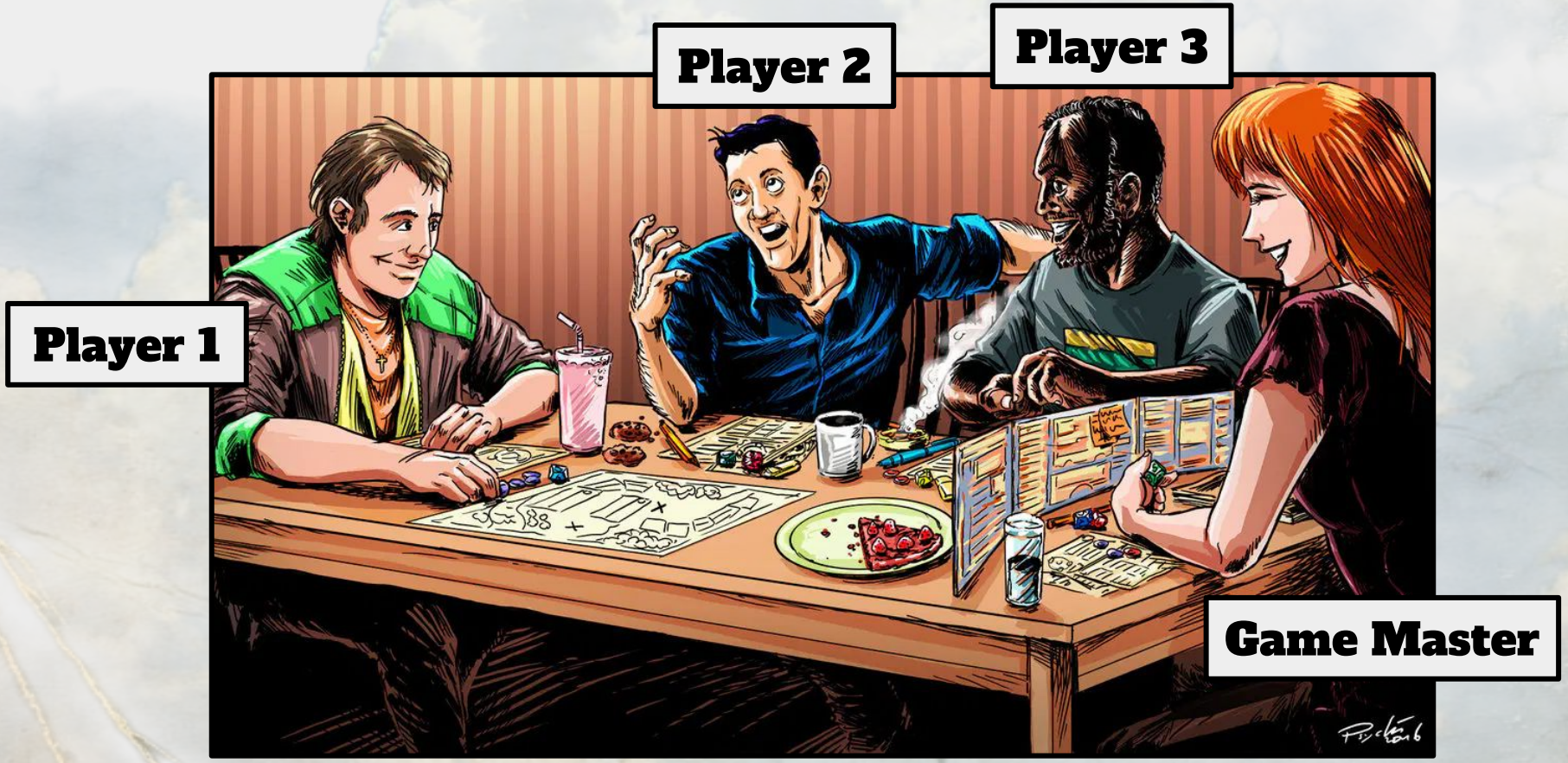
Grammars!

There are some options here to control how the generation process behaves...

Neurosymbolic text generation



Tabletop Role-playing Games (TTRPG)



AI Dungeon

AI Dungeon is a TTRPG model originally built using GPT-2, back in 2019.

As it has been receiving many updates throughout the years, this is still a very relevant TTRPG model.



LLMs can be prompted to act as Game Masters

Neural Models Can Drift Without Symbolic Control

Taken from an AIDungeon gameplay on April 15 2024

> You check your inventory.

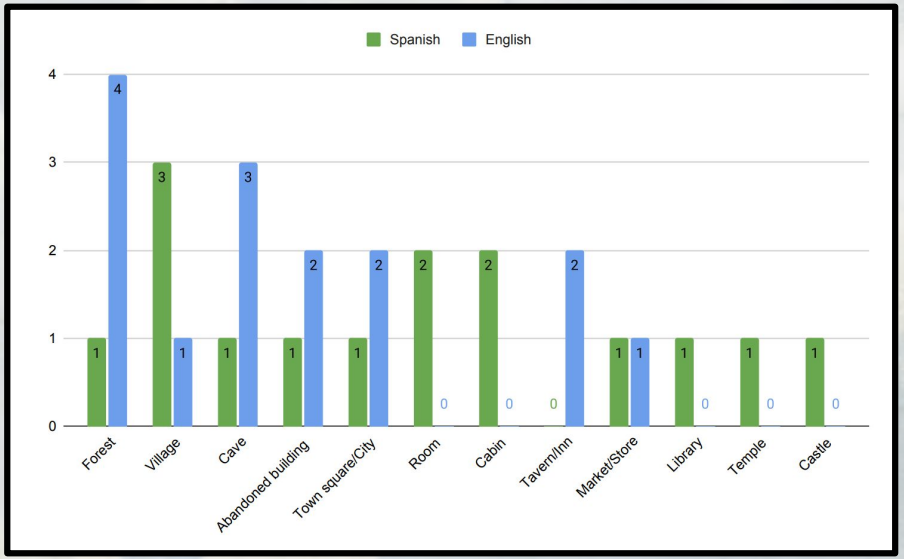
You quickly check your inventory, making sure your pistol is loaded and easily accessible. Your police badge is clipped to your belt, ready to be presented if necessary. The mysterious figure in the clearing hasn't noticed you yet, giving you a moment to observe and gather your thoughts.

> You take the bazooka from your inventory to shoot the figure.

As you reach for the bazooka in your inventory, you realize that it's not there. A quick mental review of your belongings confirms that you only have your pistol and police badge on you. Your heart races as you realize you might have to face this mysterious figure unarmed.

> You take your bazooka from the right pocket of your bag.

Upon further investigation, you find that your bazooka is indeed in your bag, specifically in the right pocket. You quickly and quietly equip it, ensuring that it's



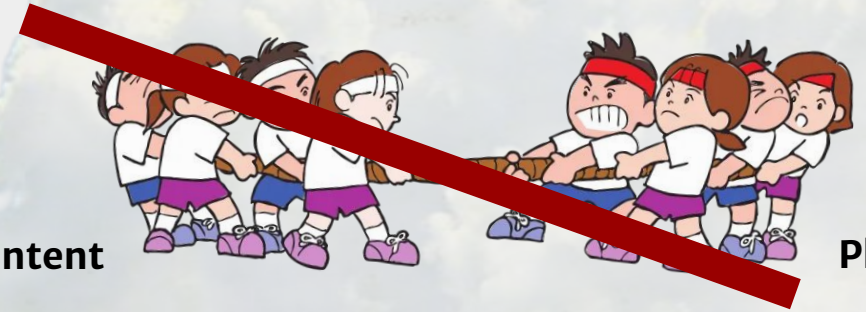
Skill Check: Some Considerations on the Evaluation of Gamemastering Models for Role-playing Games

Santiago Góngora[1[0000−0002−6256−7577]], Luis Chiruzzo[1[0000−0002−1697−4614]], Gonzalo Méndez[2[0000−0001−7659−1482]], and Pablo Gervás[2[0000−0003−4906−9837]]

[1] Universidad de la República, Uruguay {sgongora,luischir}@fing.edu.uy
[2] Universidad Complutense de Madrid, Spain {gmendez,pgervas}@ucm.es

The players choose from rigid, pre-authored options.
No room for *player expression*.

The story lacks suspense and drama.
Everything seems too easy.



Authorial Intent

Player Agency

The narrative paradox

Is it possible for LLMs to **adhere to certain narrative constraints** while still allowing **players to shape the story**, as long as the changes are valid?

PAYADOR

A PlAYable Approach based on Descriptions for Outcomes in Role-playing games

A **Neuro-symbolic** approach to model a Game Master

PAYADOR: A Minimalist Approach to Grounding Language Models on Structured Data for Interactive Storytelling and Role-playing Games
(Góngora, S., Chiruzzo, L., Méndez, G., Gervás, P., 2024)

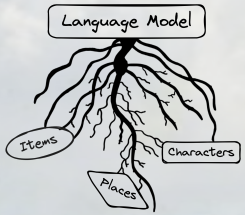
Main characteristics of PAYADOR

A structured representation

```
Name: "Catyl"
Inventory: ["Sword", "Apple"]
Location: "Hill"
Descriptions:
["She is a mage",
"She is tall",
"She knows how to cast lightning
bolts",
"Since she was a little girl, she
always loved climbing mountains"]
```

+ World-state Transformations

- Pre-programmed world-updates
- Pre-programmed logic checks

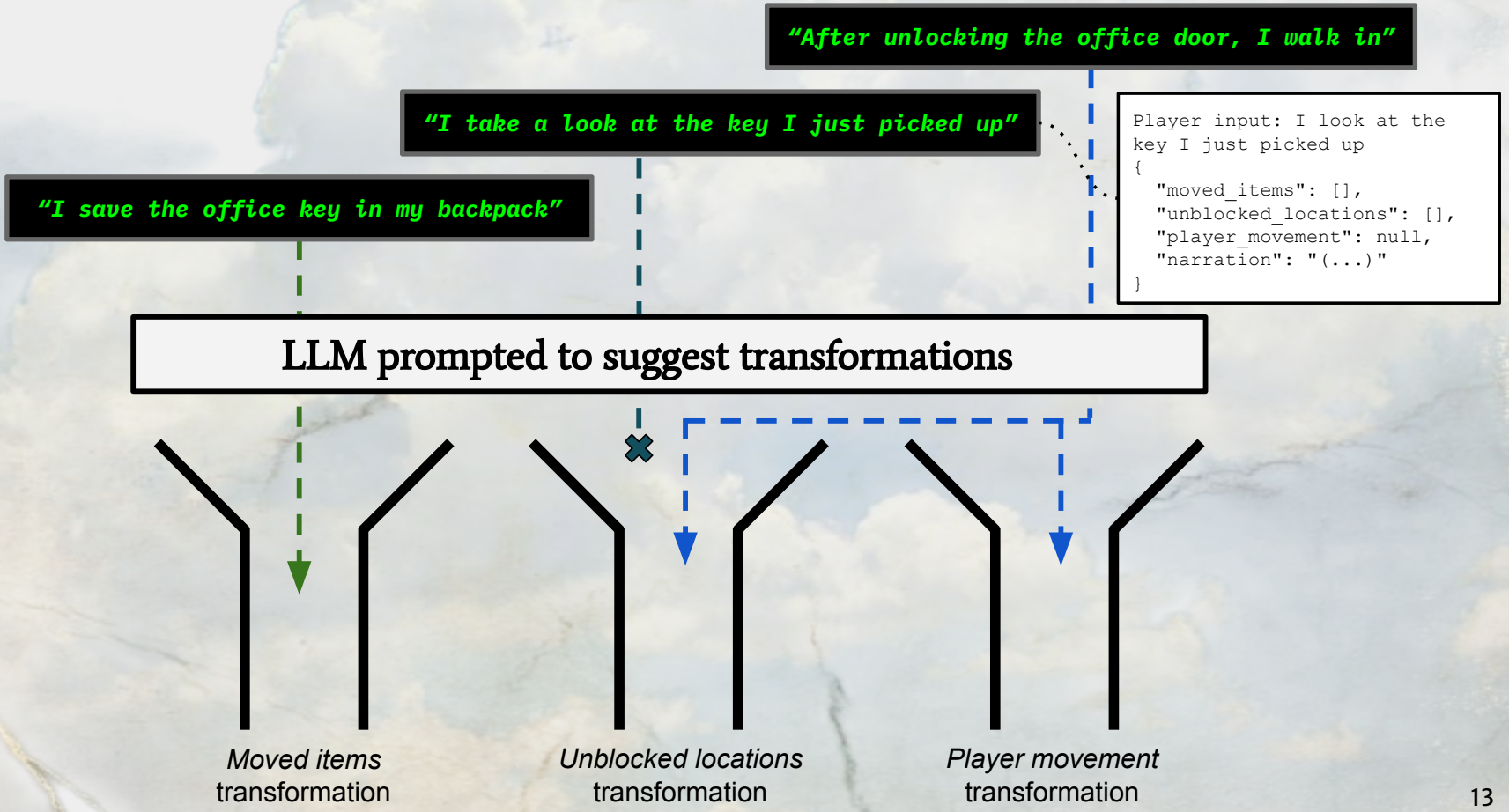


(I) A player enters **free-text input** →

(II) an LLM suggests <u>which</u> **transformations** are triggered →

(III) the symbolic modules execute consistency checks + world updates

World-State Transformations (an example)



Designing a New World-State Transformation

Which aspect of the world state may be affected (by character actions or other events)?

Every character should have one of these moods: “Happy,” “Sad,” “Angry,” or “Neutral.”



Does the structured world representation need any additional components?

```
character.mood = “Neutral”
```

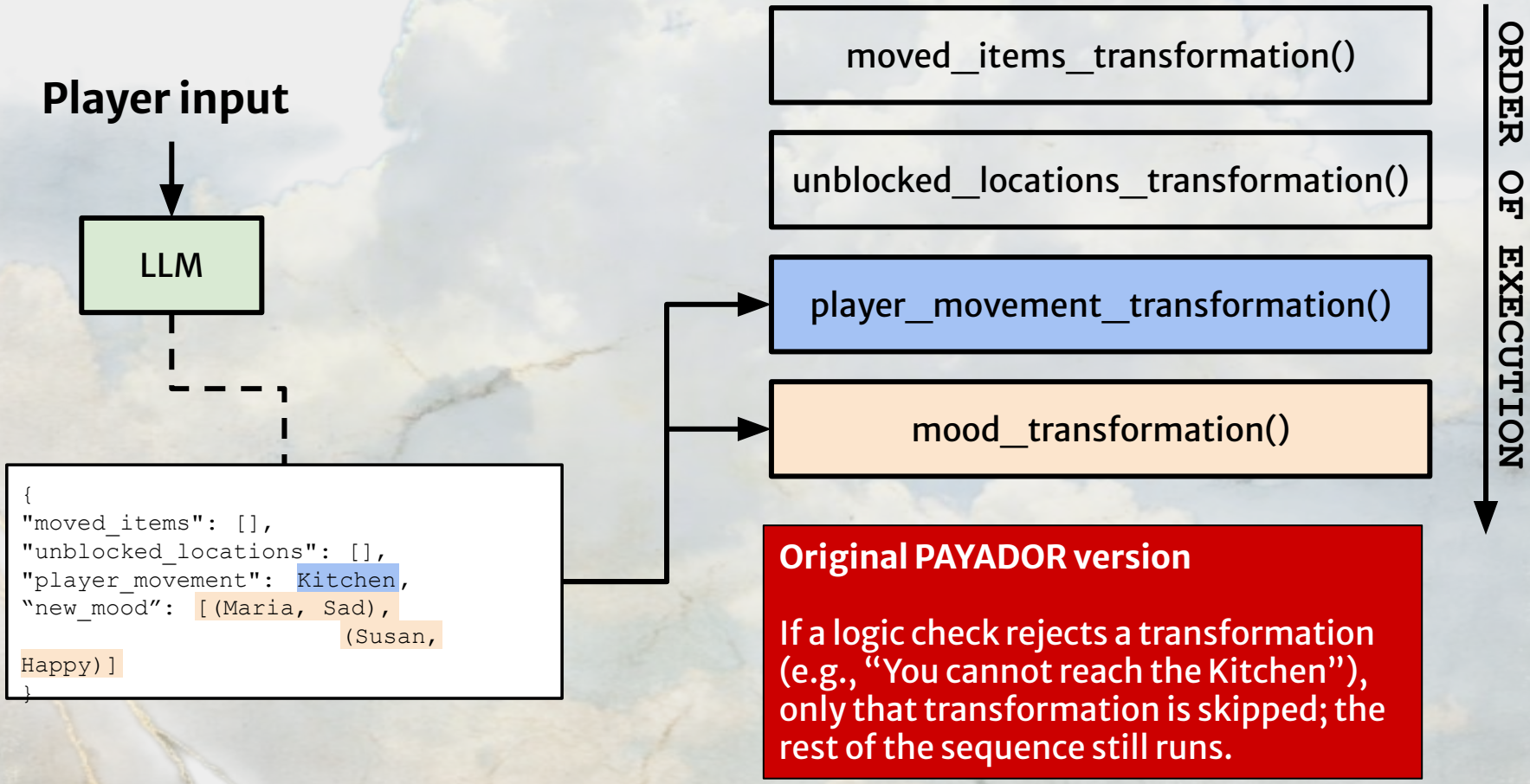
What logic checks should be performed before updating the world state (i.e., preconditions)?

```
def valid_mood_update (self,new_mood):
      (...)
      if new_mood in [“Happy”, “Sad”, “Angry”,
“Neutral”]:
      (...)
#any new_mood can come after the current
mood?
```

How is the world state actually updated?

```
def mood_transformation (self,new_mood):
      if valid_mood_update (self.mood, new_mood):
            self.mood = new_mood
```

Our ✨New ✨ World-State Transformation



Why using World-State Transformations?

GM: *At the top of the tree, you can see a bird holding a key in its beak. The key looks very similar to the one you're looking for...*

Player: *Great! So…I will take the key.*

GM: *OK… but how will you do it?*



I climb the tree to get closer to the bird. Then, I try to take the key.

I take the fishing pole I have in the garage. Then, I use it to try to get the key with it.

Why using World-State Transform

GM: *At the top of the tree, you can see a bird holding a key in its beak. The key looks very similar to the one you're looking for...*

Player: *Great! So…I will take the key.*

GM: *OK… but how will you do it?*

I put breadcrumbs on a tray to attract the bird and, when it gets closer, I take the key

(yes, this was generated with Gemini Pro)----------



Players often come up with ideas the GM never have thought of

Transformations try to model how GMs embrace players' creativity

The focus is on the effect of the actions, not on the action themselves

1

Render the world state

grounding



World state

grounding

2 A

Generate the prompt and describe the new scene

(if the character moved to a new location)

LLM

OR

2 B

Narrate the outcomes

(if the character stayed in the same location)

User input



3

Generate the prompt and predict the effects of the user input

LLM

4

Extract transformations and narrated outcomes from the predicted effects

5

Update the world state

The LLM does not have access to previous turns in the conversation



Yeah... but it's not
all good.
Tell them about it.

Limitations

Evaluation

We evaluated PAYADOR with **8 participants**, some playing in **English** and some in **Spanish**.

Each participant played two scenarios:

- a **realistic one**, to check whether world-state transformations could handle real-world physics
- a **fantasy one**, to check how the system handled non-realistic world-state manipulation

World-State Transformations for Neuro-symbolic Interactive Storytelling

Santiago Góngora[1] † **Luis Chiruzzo**[1] **Gonzalo Méndez**[2] **Pablo Gervás**[2]

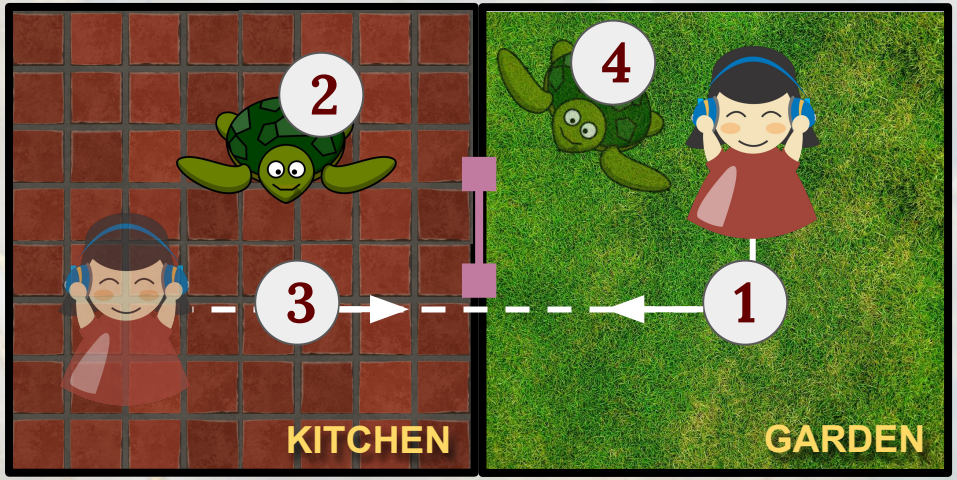
[1] Instituto de Computación, Facultad de Ingeniería, Universidad de la República, Uruguay
[2] Facultad de Informática, Universidad Complutense de Madrid, Spain
† sgongora@fing.edu.uy

Problem 1: Order of execution

Transformations were always executed in a previously-defined order

Moved Items → Unblocked Locations → Player Movement

```
"I'm going to the kitchen to get the Turtle and
then drop it on the Garden's floor"
```



1. **Go to the *Kitchen***
   (Player Movement)
2. **Pick up the *Turtle***
   (Moved Items)
3. **Go to the *Garden***
   (Player Movement)
4. **Drop the *Turtle***
   (Moved Items)

We committed a fix to the GitHub repo

However, we have not formally tested it (with humans) yet.

The LLM is also prompted to include a **sequence number** associated with each transformation.

The **symbolic world-update process** executes each transformation in the **indicated order**.

```
{
"moved_items": [],
"unblocked_locations": [],
"Player_movement": [
      {place: Kitchen,
      number: 2}]
“new_mood”: [
      {character: Maria,
      mood: Sad,
      number: 1},
      {character: Susan,
      mood: Happy,
      number: 3}]
}
```

Problem 2: When and How to Narrate Action Outcomes

Player input → LLM →  → Use the generated narration to describe the outcomes to the player

Update world-state

The original PAYADOR version asks the LLM to generate a natural-language description alongside its suggested transformations

```
Player input: I use the key I just picked
up to open the storehouse.
{
  "moved_items": [],
  "unblocked_locations": ["Storehouse"],
  "player_movement": null,
  "narration": "You use the key to open
the storehouse"
}
```

```
Error: "Storehouse" is not a valid location.
```

What if the symbolic world-state update process **rejects the suggested transformations**?

This problem is analogous to the one affecting pure-LLM TTRPG models

By trusting the LLM's output, we expose the neurosymbolic system to the typical neural problems of narrative coherence.

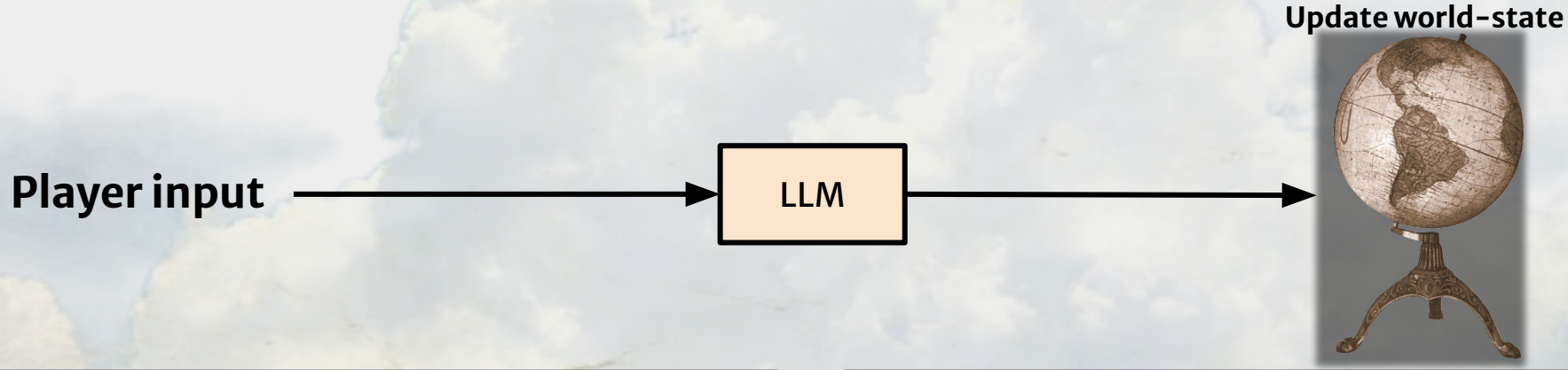
```
Player input: I use the key I just picked up to
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{
  "moved_items": [],
  "unblocked_locations": [“Storehouse”],
  "player movement": null,
  "narration": “You use the key to open the
storehouse"
}
```



> You take your bazooka from the right pocket of your bag.

Upon further investigation, you find that your bazooka is indeed in your bag, specifically in the right pocket. You quickly and quietly equip it, ensuring that it's

Problem 2: When and How to Narrate Action Outcomes



Option 1

The narration is generated as in the original version, but in two variants: **a positive** one and **a negative** one. If something goes wrong, all the transformations are reverted to the original world state, and the player reads the negative version of the narration.

Option 2

The narration is generated **after the symbolic world-update process**, based on the difference between the new and the original world states:

diff(NewWS,OriginalWS)

We committed a fix to the GitHub repo

However, we have not formally tested it (with humans) yet.

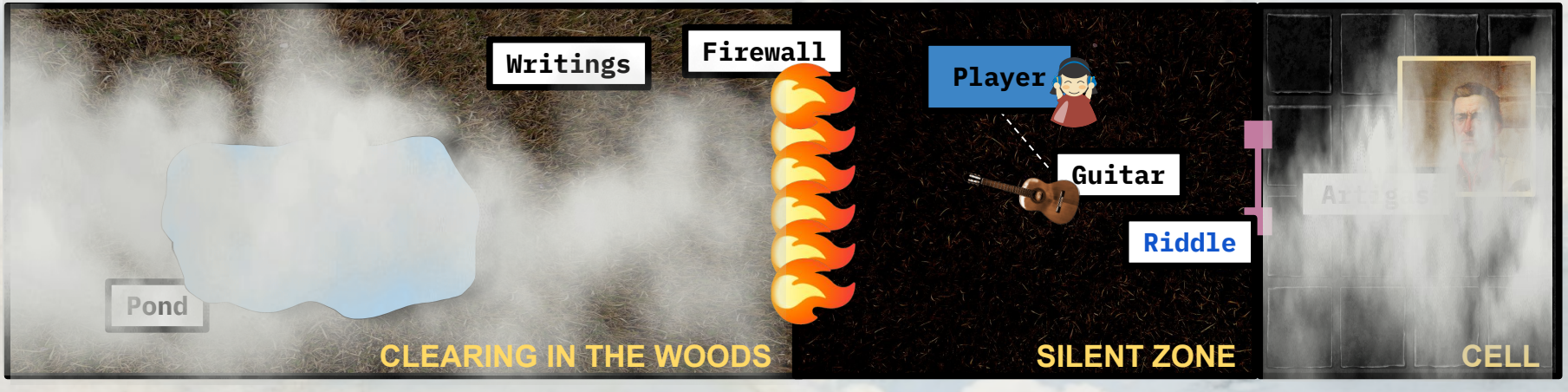
Option 1

The narration is generated as in the original version, but in two variants: **a positive** one and **a negative** one.

If something goes wrong, all the transformations are reverted to the original world state, and the player reads the negative version of the narration.

```
Player input: I use the key I just picked up to
open the storehouse.
{
  "moved_items": [],
  "unblocked_locations": [“Storehouse”],
  "player movement": null,
  "positive_narration": “You use the key to open
the storehouse"
  "negative_narration": “The door does not budge."
}
```

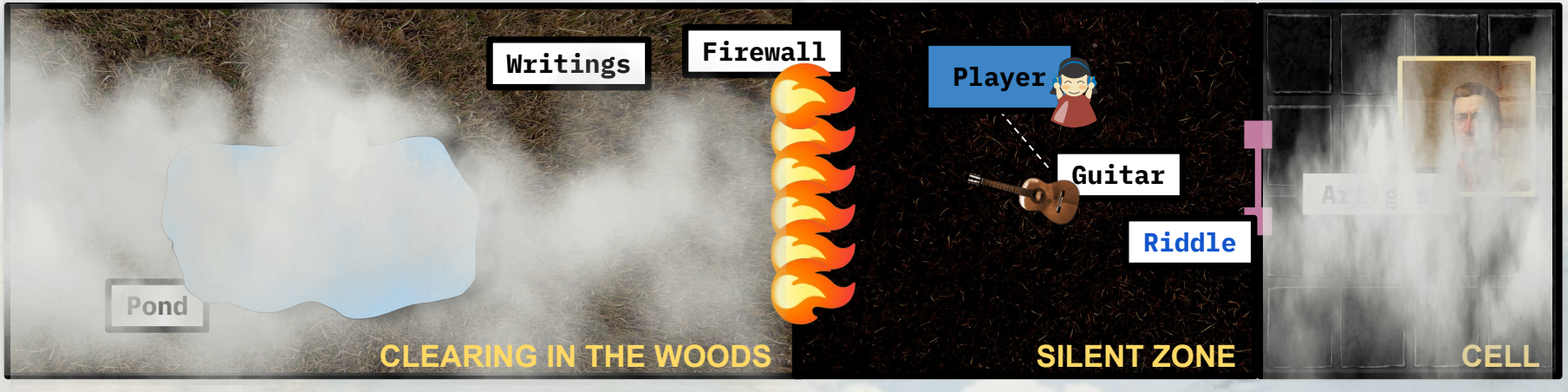
Problem 3: What Should the LLM Know About the Symbolic World State at Each Step?



Currently, the LLM gets the world state following a *fog of war* policy.

This restricts the LLM to answer questions about components in adjacent locations, and to generate interesting scene descriptions.

Problem 3: What Should the LLM Know About the Symbolic World State at Each Step?



Should the LLM receive the world state of adjacent areas?

Should the LLM receive the entire global world state?

Problem 4: Neuro-symbolic memory

Intentionally, PAYADOR relies only on the aspects saved in the symbolic world state. **Previous utterances are not persisted.**

Turn 8

GM (Gemini Flash 1.5): "(...) What do you want to do? You can examine the objects in the art studio, talk to Laura, or try to go to the kitchen"

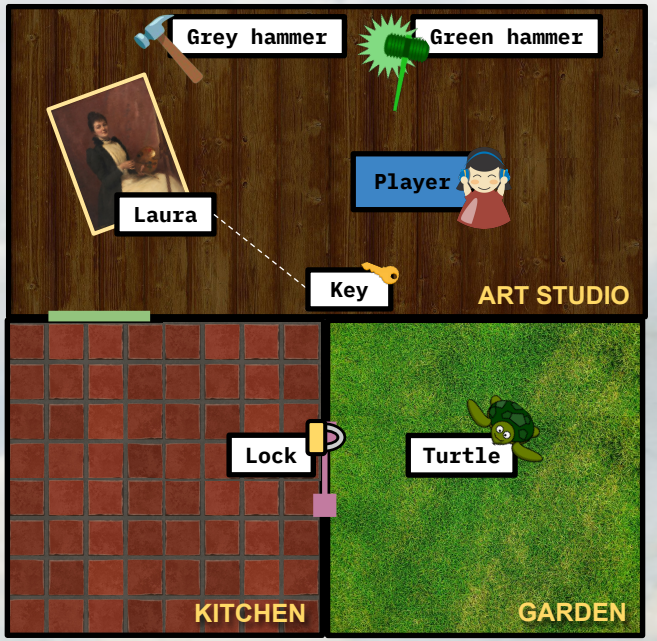
Player: "I look at the green hammer, could it be hojita?"

Turn 9

GM (Gemini Flash 1.5): "You look at the green hammer. It's small, plastic, and clearly a toy. It doesn't seem to have any connection to 'Hojita'. Would you like to try something else?"

Player: "I grab the other hammer and head to the kitchen"

Actual gameplay. Translated from the original Spanish.

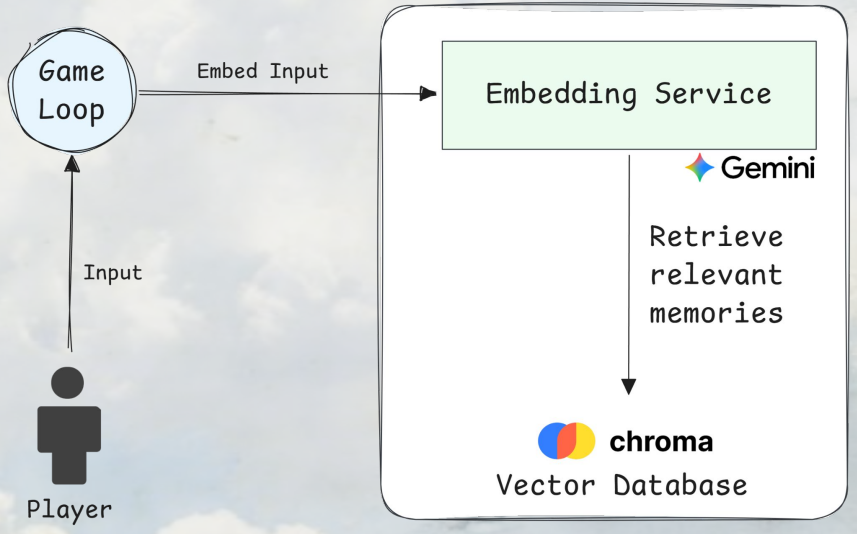



```
- Moved object: <Un martillo verde> now is in <Inventory>
- Blocked passages now available: None
- Your location changed: <Cocina>
```

RAG seemed to work... (at first)

IVIE is a world generator for PAYADOR, built directly on top of it.

This system uses RAG to overcome PAYADOR's memory problems.



Although it received positive feedback, human evaluation showed that state-of-the-art LLMs **still mix up details within the context**.

Can we trust SOTA LLMs for every interactive storytelling task?

Logic checks

Symbolic memory

How can we find the optimal boundary between symbolic approaches and LLMs?

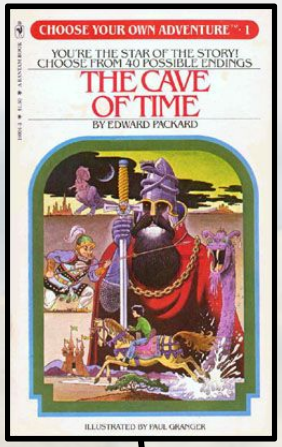
Many research questions!

Beyond measurable performance, are there other, **less obvious reasons** to pursue neurosymbolic approaches?

How can we evaluate all of this? Is there an automatic metric yet to be found?

These are just a few of them :)

The Narrative Paradox Is (Definitely) Still Alive



This is also a **known problem** in human-played TTRPGs



Railroading

Excessively **rigid** narrative structure



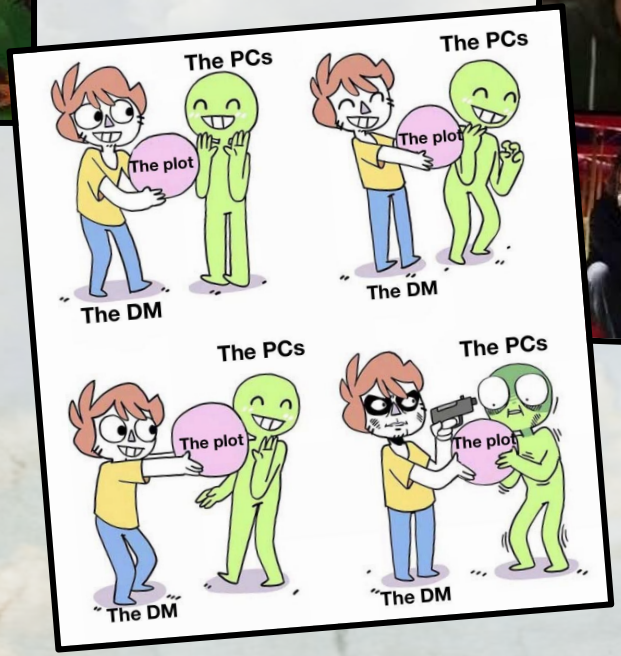
Excessively **loose** narrative structure

I would place PAYADOR around here



When the players aren't following the story like you imagined they would







The Narrative Paradox is Still Alive

Is the Narrative Paradox Immortal?

Is the Narrative Paradox a Law of Human Nature?

Finding the right balance is an artistic choice

It's more complex than a regular problem of subjectivity in NLP:
many subjective choices evaluated by **many subjective perceptions**

How should the author's views be represented in the story?

What should the story provoke in the players?

What do the players want, and what narrative decisions are needed to deliver it?

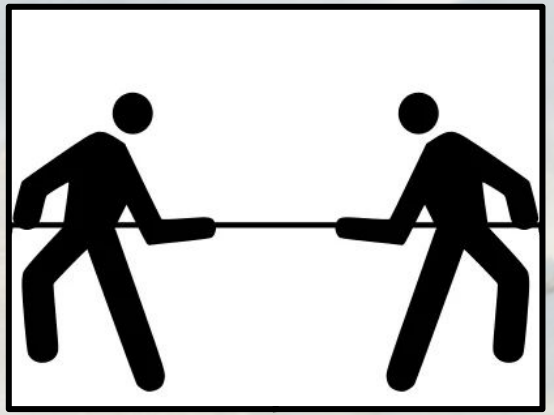
How much does it matter if the story deviates from the initial plan?

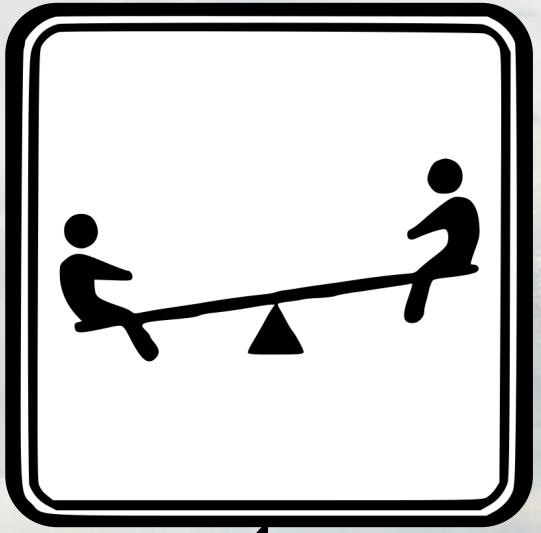
Original question: Is it possible for LLMs to **adhere to certain narrative constraints** while still allowing **players to shape the story**, as long as the changes are valid?

New question: Can LLMs co-create stories with the same **imperfect magic** as humans do?

How might symbolic components help achieve this goal?

I believe we can do better.





Thank you!

Questions? :)